

Period Intensity Adjustment

A Bayesian Personalisation Model for Menstrual-Phase Training Load

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Abstract

Fluctuations in oestrogen and progesterone across the menstrual cycle measurably affect perceived exertion, thermoregulation, substrate utilisation, and injury risk. A fixed training prescription that ignores this variation is sub-optimal for female athletes. This paper describes the model implemented in StravaEat for *personalised* intensity reduction during menstruation. The system begins with conservative clinical heuristics as starting priors and refines them through a Bayesian conjugate update each time the athlete provides post-session feedback on a 4-point perceived-effort scale. Because the posterior is dominated by individual data after only a few sessions, the exact choice of prior matters little — convergence to the athlete's true tolerance is guaranteed regardless of the starting values.

1. Physiological Background

The menstrual cycle is conventionally divided into the **follicular phase** (days 1–13, characterised by rising oestrogen) and the **luteal phase** (days 15–28, characterised by high progesterone and elevated resting core temperature). Menstruation itself (days 1–5) is associated with prostaglandin-mediated uterine contractions, systemic inflammation, and fatigue that is independent of haemoglobin loss in most athletes.

McNulty et al. (2020) conducted a systematic review and network meta-analysis of 78 studies (51 in the primary meta-analysis, 73 in the network meta-analysis) and found that exercise performance is **trivially reduced during the early follicular phase** (i.e. menstruation) compared with all other cycle phases ($ES = -0.14$, 95% credible interval $[-0.26, -0.03]$). The early follicular phase had the lowest SUCRA ranking (30%) versus 53–55% for all other phases. Overall evidence quality was rated low (42%). The authors explicitly recommend a **personalised approach** due to the trivial average effect and large between-individual variability, which motivates the adaptive Bayesian model implemented here.

The literature supports the following zone-specific adjustments during menstruation:

- **Zone 2 (aerobic):** Heuristic starting point: minor –20 %; moderate –40 %
- **Zone 3 (tempo):** Heuristic starting point: minor –30 %; moderate –50 %
- **Zone 4/5 (threshold / VO_{2max}):** Minor: reduce to Zone 3; Moderate: replace entirely with Zone 2

The StravaEat model captures this with a single scalar **reduction fraction** $r \in (0, 1)$ representing the proportional decrease in target training load (kcal, duration, intensity). Zone-specific modifiers are layered on top as fixed offsets.

2. Prior Specification

For each severity level $s \in \{\text{minor, moderate}\}$ we specify a Gaussian prior over the reduction fraction r :

$$r \mid s \sim \mathcal{N}(\mu_0(s), \sigma_0^2(s))$$

Prior distribution over the reduction fraction r

The prior parameters are **conservative clinical heuristics chosen as starting points**, not evidence-based values derived from the literature. McNulty et al. (2020) found only trivial average effects ($ES = -0.14$) and explicitly advise against general load-reduction guidelines. The 20 % and 40 % means reflect cautious upper-bound estimates from coaching practice; the model is designed to move away from them rapidly as individual feedback accumulates. As shown in Section 7, the prior is outweighed by the athlete's own data after just three sessions — the exact starting values matter little.

Severity	Prior mean μ_0	Prior std σ_0	Prior variance σ_0^2	Source
Minor	0.20 (20 %)	0.10 (10 %)	0.0100	Conservative coaching heuristic — upper-bound starting point for minor symptoms
Moderate	0.40 (40 %)	0.12 (12 %)	0.0144	Conservative coaching heuristic — upper-bound starting point for moderate symptoms

Table 1. Population prior parameters.

The σ_0 values (0.10 for minor, 0.12 for moderate) are chosen to reflect meaningful but not excessive uncertainty around the starting means. The wider prior for moderate symptoms acknowledges greater variability in self-reported symptom severity at that level. Both values are intentionally broad so that athlete feedback quickly dominates the posterior — see Section 7 for convergence rates.

Severe symptoms are treated as a qualitatively different category: the model prescribes complete rest or gentle movement only, and no Bayesian update is performed. Severe is therefore excluded from the continuous calibration framework.

3. Observation Model

After each training session completed during menstruation the athlete provides feedback on a 4-point perceived-effort scale:

Rating	Label	Implied signal x
1	Much too easy	$x = r_{\text{prescribed}} - 0.10$
2	Slightly easy	$x = r_{\text{prescribed}} - 0.04$
3	Slightly hard	$x = r_{\text{prescribed}} + 0.04$
4	Too hard	$x = r_{\text{prescribed}} + 0.10$

Table 2. Feedback ratings and implied observation signals.

The observation signal x is modelled as the true reduction fraction r plus Gaussian noise:

$$X = r + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_{\text{obs}}^2)$$

Observation model

The noise variance $\sigma_{\text{obs}}^2 = 0.07^2 = 0.0049$ is derived from the test-retest reliability of subjective effort scales (Borg 1970; Noble et al. 1983), which exhibit an intra-subject coefficient of variation of approximately 7 % for session RPE.

The asymmetric step sizes $\Delta \in \{-0.10, -0.04, +0.04, +0.10\}$ are calibrated such that the outer steps (ratings 1 and 4) equal approximately one prior standard deviation σ_0 , producing a full Bayesian shift for a strong signal, while the inner steps (2 and 3) produce a gentle $0.4\sigma_0$ nudge for a weak signal. This prevents the model from over-reacting to ambiguous feedback.

4. Bayesian Conjugate Update

Under the Gaussian likelihood and Gaussian prior the posterior is also Gaussian — the conjugate pair. Given the current posterior (μ_n, σ_n^2) and a new observation x , the updated posterior is:

$$\frac{1}{\sigma_{n+1}^2} = \frac{1}{\sigma_n^2} + \frac{1}{\sigma_{\text{obs}}^2}$$

Posterior precision update (precision = 1 / variance)

$$\mu_{n+1} = \sigma_{n+1}^2 \cdot \left(\frac{\mu_n}{\sigma_n^2} + \frac{x}{\sigma_{\text{obs}}^2} \right)$$

Posterior mean update (precision-weighted average)

This is the standard Bayesian update for a known-variance Gaussian model (Murphy 2007, §7.6). Three properties are highlighted:

- **Precision accumulates.** Each observation adds $1/\sigma^2_{\text{obs}} \approx 204$ units of precision. After three observations the data outweigh the prior, and the posterior mean is dominated by the athlete's own feedback.
- **Posterior variance is monotonically decreasing.** $\sigma^2_{n+1} < \sigma^2_n$ for all n , so the model becomes more confident with every session.
- **The update is commutative.** The order of observations does not affect the final posterior; only the number of them matters.

The closed-form posterior variance after n observations starting from prior (μ_0, σ^2_0) is:

$$\sigma_n^2 = \frac{1}{\frac{1}{\sigma_0^2} + \frac{n}{\sigma_{\text{obs}}^2}}$$

Closed-form posterior variance after n i.i.d. observations

5. Posterior Clipping

To prevent the posterior mean from drifting to physiologically degenerate values it is clipped to a feasible interval after each update:

$$\mu_{n+1} \leftarrow \text{clip}(\mu_{n+1}, 0.05, 0.70) = \max(0.05, \min(0.70, \mu_{n+1}))$$

Clipping to the feasible interval $[0.05, 0.70]$

The lower bound $\mu_{\text{min}} = 0.05$ prevents the model from concluding that the athlete requires zero reduction (contradicting the physiological literature). The upper bound $\mu_{\text{max}} = 0.70$ prevents prescribing more than 70 % load reduction, which is reserved for the severe category. Clipping only affects the mean; the variance update is not clipped and continues to decrease normally.

6. Zone-Specific Load Application

The posterior mean μ_n is the primary reduction fraction applied to the overall training load. Zone-specific modifiers Δ_z are layered additively:

$$r_z = \min(1, \mu_n + \Delta_z)$$

Effective reduction fraction for heart rate zone z

Zone	Description	Δ_z	Rationale
Z1	Recovery	0	Recovery load is already minimal
Z2	Aerobic	0	Base reduction μ_n applied directly
Z3	Tempo	+0.10	Higher prostaglandin sensitivity at moderate-high intensity
Z4	Threshold	+0.20	Near-threshold work is disproportionately affected by luteal-phase fatigue
Z5	VO ₂ max	+0.20	Replace with Z3; VO ₂ max intervals not recommended during menstruation

Table 3. Zone-specific reduction offsets Δ_z .

Applied to the planned workout, the adjusted energy target and duration are:

$$E_{\text{adj}} = E_{\text{planned}} \cdot (1 - r_z)$$

Adjusted energy target

$$t_{\text{adj}} = t_{\text{planned}} \cdot (1 - \mu_n)$$

Adjusted duration (base reduction only, no zone offset)

7. Convergence Analysis

The calibration confidence displayed in the UI is defined as the fractional reduction in posterior variance relative to the prior variance:

$$\text{confidence}(n) = 1 - \frac{\sigma_n^2}{\sigma_0^2} = \frac{n/\sigma_{\text{obs}}^2}{1/\sigma_0^2 + n/\sigma_{\text{obs}}^2}$$

Calibration confidence: fraction of prior variance explained by observations

For the minor severity prior ($\sigma_0^2 = 0.01$, $\sigma_{\text{obs}}^2 = 0.0049$), the confidence after n sessions is:

Sessions (n)	σ_n^2	σ_n	confidence(n)	UI label
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0	0.0100	0.100	0 %	Population default
1	0.0033	0.057	67 %	Early learning
2	0.0021	0.046	79 %	Early learning
3	0.0015	0.039	85 %	Personalising
5	0.0009	0.030	91 %	Personalising
6	0.0008	0.028	92 %	Personalised
10	0.0005	0.022	95 %	Personalised

Table 4. Convergence of the posterior for the minor severity prior.

The model reaches “Personalised” status (≥ 92 % confidence) after approximately **6 sessions**, corresponding to roughly 2 menstrual cycles at 3 training sessions per cycle. From session 1 the posterior already incorporates the individual’s data; convergence is gradual and monotonic.

Prior insensitivity. A key property of Bayesian updating is that the posterior converges to the athlete's true reduction tolerance regardless of the starting prior, provided the prior has support over the true value and enough observations accumulate. The prior affects only the speed of early convergence, not the eventual destination. Because $\sigma_{\text{obs}}^2 = 0.0049$ is small relative to both σ_0^2 values, each feedback session contributes substantial information and the influence of the starting means (20 % and 40 %) fades quickly. An athlete whose true tolerance is 10 % will converge there just as reliably as one whose true tolerance is 35 % — the model makes no long-run commitment to its clinical heuristic starting points.

8. Calendar Example: Two Athletes

To make convergence concrete, consider two athletes both training four days per week (Monday and Thursday), with a 28-day cycle and five-day menstruation window. On average $5/7 \times 4 \approx 2.9$ training days fall within menstruation per cycle; with a Mon/Thu schedule this rounds to **2 sessions per cycle**.

Athlete A has *minor* symptoms. The app starts at the population default of 20 % reduction. Her true personal tolerance turns out to be approximately 14 % — the prior overshoots her, so early sessions feel too easy.

Athlete B has *moderate* symptoms. The app starts at 40 % reduction. Her true tolerance is approximately 43 % — the prior slightly undershoots, so early sessions feel marginally hard.

8.1 Athlete A — minor severity, true tolerance ≈ 14 %

Session	Date	Cycle	Prescribed	Feedback	μ after	Confidence	Status
1	Mon 3 Mar	1	20.0 %	Much too easy (1)	13.3 %	67 %	Early learning
2	Thu 6 Mar	1	13.3 %	Slightly hard (3)	14.9 %	80 %	Early learning

3	Mon 31 Mar	2	14.9 %	Slightly easy (2)	13.7 %	86 %	Personalising
4	Thu 3 Apr	2	13.7 %	Slightly hard (3)	14.6 %	89 %	Personalising
5	Mon 28 Apr	3	14.6 %	Slightly easy (2)	13.9 %	91 %	Personalising
6	Thu 1 May	3	13.9 %	Slightly hard (3)	14.5 %	92 %	Personalised ✓

Table 5. Athlete A convergence. Session 1 sees the largest single correction (−6.7 pp) as the prior default of 20 % is well above her true tolerance. The model settles around 14 % by session 3 and is stable from session 6 onward. Calendar time: approximately 2 months (3 Mar – 1 May).

8.2 Athlete B — moderate severity, true tolerance $\approx$ 43 %

Session	Date	Cycle	Prescribed	Feedback	μ after	Confidence	Status
1	Mon 3 Mar	1	40.0 %	Slightly hard (3)	43.0 %	75 %	Early learning
2	Thu 6 Mar	1	43.0 %	Slightly hard (3)	44.7 %	86 %	Early learning
3	Mon 31 Mar	2	44.7 %	Slightly easy (2)	43.5 %	90 %	Personalising
4	Thu 3 Apr	2	43.5 %	Slightly easy (2)	42.6 %	92 %	Personalised ✓
5	Mon 28 Apr	3	42.6 %	Slightly hard (3)	43.3 %	94 %	Personalised
6	Thu 1 May	3	43.3 %	Slightly easy (2)	42.7 %	95 %	Personalised

Table 6. Athlete B convergence. Because the moderate prior is wider ($\sigma^2 = 0.0144$ vs 0.0100) and the true value is close to the prior mean, each session carries more relative weight. Personalised status is reached at session 4 — end of cycle 2, roughly 5 weeks of calendar time (3 Mar – 3 Apr).

Two observations stand out. First, **the prior correction in session 1 is asymmetric**: Athlete A drops from 20 % to 13.3 % in a single session (a −6.7 pp jump) because the prior was far from her truth; Athlete B barely moves (40 % → 43.0 %) because the prior was already close. Second, **Athlete B converges faster in calendar time** — cycle 2 versus cycle 3 — not because she trains more, but because the wider moderate prior discounts itself more readily with each observation, consistent with the prior-insensitivity property discussed in Section 7.

8. Implementation

8.1 Data model

The calibration state is persisted in the **period_calibration** table:

```
period_calibration(
  user_id      uuid,    -- FK → auth.users
  severity     text,    -- "minor" | "medium"
  mu           float8,  -- posterior mean
```

```

sigma_sq      float8, -- posterior variance
n_obs         integer, -- observation count
last_feedback_date date, -- one update per day guard
UNIQUE (user_id, severity)
)

```

8.2 Update function (TypeScript)

The core update is a pure function in **mobile/lib/periodCalibration.ts**:

```

function bayesianUpdate(
  mu: number,
  sigma_sq: number,
  prescribedReduction: number,
  rating: 1 | 2 | 3 | 4
): { mu: number; sigma_sq: number } {
  const x = prescribedReduction + FEEDBACK_DELTAS[rating]
  const new_sig_sq = 1 / (1/sigma_sq + 1/SIGMA_OBS_SQ)
  const new_mu = new_sig_sq * (mu/sigma_sq + x/SIGMA_OBS_SQ)
  return { mu: clamp(new_mu, MU_MIN, MU_MAX), sigma_sq: new_sig_sq }
}

```

8.3 Feedback guard

The feedback prompt is shown at most once per calendar day and only after a completed activity:

```

periodActive          // sex === "female" && onPeriod
&& periodSeverity !== "severe"
&& todayActivities.length > 0
&& periodCalib?.last_feedback_date !== localDate()

```

9. Limitations and Future Work

- **Single-session feedback granularity.** The 4-point scale conflates multiple fatigue sources (sleep, nutrition, stress). A richer signal — e.g. HRV or a multi-dimensional post-session questionnaire — would reduce σ^2_{obs} and accelerate convergence.
- **Cycle-phase independence assumption.** The model assumes a fixed per-menstruation effect and does not model intra-cycle variation. A future version could maintain separate posteriors per phase using a hierarchical model.
- **No forgetting mechanism.** The posterior variance decreases monotonically. If physiology changes (e.g. due to contraceptive use), old observations are never discounted. An exponential-forgetting variant would address this.
- **i.i.d. assumption.** Consecutive cycles are correlated and within-cycle fatigue accumulates. A state-space or autoregressive model would be more appropriate for longer time series.
- **Severe severity excluded.** Athletes with dysmenorrhoea or endometriosis may have highly variable severity. A continuous severity dimension rather than three discrete categories is a natural extension.

Despite these simplifications the Bayesian Gaussian model is well-suited to a proof-of-concept implementation: it is analytically tractable, interpretable, requires no

training data, and produces well-calibrated uncertainty estimates that can be displayed directly to the user.

References

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This document describes a proof-of-concept model. Parameters should be reviewed against emerging literature. The system is not a medical device and does not constitute medical advice.